ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 87	$\frac{12\times 89+63}{13}=87$.
2	**(d)** 85	$\frac{27\times 84-58}{26}=85$.
3	**(a)** 64	Sum of the extra numbers $=70\times 60-40\times 73=1280$; divide by 20.
4	**(a)** 24	$12(73-71)=n_2(74-73)$.
5	**(b)** 18%	$\frac{360\times 35+680\times 9}{1040}=18\%$.
6	**(a)** 74	$\frac{62\times 3+78\times 4+79\times 4}{11}=74$.
7	**(a)** 70	New salary is higher by $n\times 3=42$ thousand.
8	**(a)** 15%	$\frac{3000\times 9+2000\times 24}{5000}=15\%$.
9	**(d)** 85	$\frac{37\times 86+48}{38}=85$.
10	**(a)** 63	$\frac{31\times 66-156}{30}=63$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $=32\times 44=1408$; Remaining sum $=1408-106=1302$ (1); New mean $=\frac{1302}{31}=42$ (1).	1+1
12	Rise in total $=18\times 2=36$ thousand; New salary = old salary + rise in total (1); $=33+36=69$ thousand, i.e. ₹69000 (1).	1+1
13	Take distance each way $=168$ km, so total distance $=336$ km; Time $=\frac{168}{42}+\frac{168}{56}=7$ h (1); Average speed $=\frac{336}{7}=48$ km/h (not the simple mean of 42 and 56) (1).	1+1
14	Sum $=36\times 56=2016$; Remaining sum $=2016-126=1890$ (1); New mean $=\frac{1890}{35}=54$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the number of remaining students be n; $21\times 56+36n=41(21+n)$ (1); $n(36-41)=21(41-56)$ (1); $n=63$ (1).	1+1+1
16	Total score of all $=68\times 99=6732$; Total score of the first group $=45\times 80=3600$ (1); Total score of the additional group $=3132$ (1); Average $=\frac{3132}{54}=58$ (1).	1+1+1
17	Acid present $=\frac{36}{100}\times 180=64.8$ mL (unchanged) (1); New volume $=180+60=240$ mL (1); New concentration $=\frac{64.8}{240}\times 100=27\%$ (1).	1+1+1
	OR: Let the number of remaining students be n; $38\times 47+37n=42(38+n)$ (1); $n(37-42)=38(42-47)$ (1); $n=38$ (1).	1+1+1
18	Total score of all $=84\times 72=6048$; Total score of the first group $=32\times 74=2368$ (1); Total score of the additional group $=3680$ (1); Average $=\frac{3680}{40}=92$ (1).	1+1+1
	OR: Acid present $=\frac{11}{100}\times 160=17.6$ mL (unchanged) (1); New volume $=160+60=220$ mL (1); New concentration $=\frac{17.6}{220}\times 100=8\%$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Totals: 1886, 2200, 1920 (1); Grand total $=6006$ for 78 students (2); Average $=77$ (2).	1+2+2
	OR: Pure part in the first $=510\times\frac{22}{100}=112.2$ mL (1); Pure part in the second $=102$ mL (1); Total $=214.2$ mL in 1190 mL (1); Concentration $=18\%$ (2).	1+1+1+2

20	Totals: 1830, 1230, 2988 (1); Grand total $= 6048$ for 96 students (2); Average $= 63$ (2).	1+2+2
	OR: Pure part in the first $= 320 \times \frac{34}{100} = 108.8$ mL (1); Pure part in the second $= 208$ mL (1); Total $= 316.8$ mL in 720 mL (1); Concentration $= 44\%$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 115	Sum of the extra numbers $= 65 \times 57 - 38 \times 40 = 2185$; divide by 19.
2	**(d)** 84	$28(62 - 68) = n_2(60 - 62)$.
3	**(a)** 33%	$\frac{800 \times 12 + 700 \times 57}{1500} = 33\%$.
4	**(a)** 91	$\frac{97 \times 7 + 90 \times 6 + 79 \times 3}{16} = 91$.
5	**(c)** 93	New salary is higher by $n \times 3 = 60$ thousand.
6	**(c)** 34%	$\frac{7000 \times 39 + 5000 \times 27}{12000} = 34\%$.
7	**(b)** 74	$\frac{23 \times 75 + 51}{24} = 74$.
8	**(b)** 81	$\frac{19 \times 83 - 119}{18} = 81$.
9	**(d)** 78	Sum of the extra numbers $= 70 \times 65 - 40 \times 65 = 1950$; divide by 25.
10	**(b)** 437	$23(84 - 65) = n_2(85 - 84)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rise in total $= 25 \times 3 = 75$ thousand; New salary = old salary + rise in total (1); $= 28 + 75 = 103$ thousand, i.e. ₹103000 (1).	1+1
12	Take distance each way $= 84$ km, so total distance $= 168$ km; Time $= \frac{84}{21} + \frac{84}{28} = 7$ h (1); Average speed $= \frac{168}{7} = 24$ km/h (not the simple mean of 21 and 28) (1).	1+1
13	Sum $= 40 \times 62 = 2480$; Remaining sum $= 2480 - 101 = 2379$ (1); New mean $= \frac{2379}{39} = 61$ (1).	1+1
14	Rise in total $= 27 \times 5 = 135$ thousand; New salary = old salary + rise in total (1); $= 27 + 135 = 162$ thousand, i.e. ₹162000 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the number of remaining students be n; $11 \times 86 + 74n = 77(11 + n)$ (1); $n(74 - 77) = 11(77 - 86)$ (1); $n = 33$ (1).	1+1+1
16	Total score of all $= 45 \times 85 = 3825$; Total score of the first group $= 34 \times 63 = 2142$ (1); Total score of the additional group $= 1683$ (1); Average $= \frac{1683}{51} = 33$ (1).	1+1+1
17	Acid present $= \frac{42}{100} \times 160 = 67.2$ mL (unchanged) (1); New volume $= 160 + 80 = 240$ mL (1); New concentration $= \frac{67.2}{240} \times 100 = 28\%$ (1).	1+1+1
	OR: Let the number of remaining students be n; $17 \times 35 + 58n = 41(17 + n)$ (1); $n(58 - 41) = 17(41 - 35)$ (1); $n = 6$ (1).	1+1+1
18	Total score of all $= 64 \times 55 = 3520$; Total score of the first group $= 20 \times 85 = 1700$ (1); Total score of the additional group $= 1820$ (1); Average $= \frac{1820}{35} = 52$ (1).	1+1+1
	OR: Acid present $= \frac{24}{100} \times 70 = 16.8$ mL (unchanged) (1); New volume $= 70 + 10 = 80$ mL (1); New concentration $= \frac{16.8}{80} \times 100 = 21\%$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Totals: 3256, 3182, 2220 (1); Grand total $= 8658$ for 111 students (2); Average $= 78$ (2).	1+2+2
	OR: Pure part in the first $= 740 \times \frac{43}{100} = 318.2$ mL (1); Pure part in the second $= 114.7$ mL (1); Total $= 432.9$ mL in 1110 mL (1); Concentration $= 39\%$ (2).	1+1+1+2
20	Totals: 1071, 2520, 2784 (1); Grand total $= 6375$ for 125 students (2); Average $= 51$ (2).	1+2+2

	OR: Pure part in the first $= 850 \times \frac{22}{100} = 187$ mL (1); Pure part in the second $= 275.4$ mL (1); Total $= 462.4$ mL in 1360 mL (1); Concentration $= 34\%$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 23%	$\frac{660\times13+440\times38}{1100}=23\%$.
2	**(b)** 66	$\frac{76\times6+63\times8+54\times3}{17}=66$.
3	**(a)** 77	New salary is higher by $n\times2=38$ thousand.
4	**(d)** 24%	$\frac{6000\times21+9000\times26}{15000}=24\%$.
5	**(b)** 80	$\frac{30\times81+50}{31}=80$.
6	**(d)** 80	$\frac{15\times78-50}{14}=80$.
7	**(d)** 2	Sum of the extra numbers $=42\times62-40\times64=44$; divide by 22.
8	**(a)** 162	$36(49-40)=n_2(51-49)$.
9	**(c)** 27%	$\frac{400\times59+800\times11}{1200}=27\%$.
10	**(c)** 81	$\frac{77\times4+89\times4+79\times8}{16}=81$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take distance each way $=210$ km, so total distance $=420$ km; Time $=\frac{210}{30}+\frac{210}{42}=12$ h (1); Average speed $=\frac{420}{12}=35$ km/h (not the simple mean of 30 and 42) (1).	1+1
12	Sum $=44\times52=2288$; Remaining sum $=2288-138=2150$ (1); New mean $=\frac{2150}{43}=50$ (1).	1+1
13	Rise in total $=20\times4=80$ thousand; New salary $=$ old salary $+$ rise in total (1); $=23+80=103$ thousand, i.e. ₹103000 (1).	1+1
14	Take distance each way $=120$ km, so total distance $=240$ km; Time $=\frac{120}{40}+\frac{120}{60}=5$ h (1); Average speed $=\frac{240}{5}=48$ km/h (not the simple mean of 40 and 60) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let the number of remaining students be n; $38\times40+55n=45(38+n)$ (1); $n(55-45)=38(45-40)$ (1); $n=19$ (1).	1+1+1
16	Total score of all $=60\times76=4560$; Total score of the first group $=40\times87=3480$ (1); Total score of the additional group $=1080$ (1); Average $=\frac{1080}{36}=30$ (1).	1+1+1
17	Acid present $=\frac{19}{100}\times130=24.7$ mL (unchanged) (1); New volume $=130+60=190$ mL (1); New concentration $=\frac{24.7}{190}\times100=13\%$ (1).	1+1+1
	OR: Let the number of remaining students be n; $31\times43+52n=49(31+n)$ (1); $n(52-49)=31(49-43)$ (1); $n=62$ (1).	1+1+1
18	Total score of all $=64\times75=4800$; Total score of the first group $=25\times74=1850$ (1); Total score of the additional group $=2950$ (1); Average $=\frac{2950}{50}=59$ (1).	1+1+1
	OR: Acid present $=\frac{40}{100}\times50=20$ mL (unchanged) (1); New volume $=50+50=100$ mL (1); New concentration $=\frac{20}{100}\times100=20\%$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Totals: $3330, 2736, 3034$ (1); Grand total $=9100$ for 130 students (2); Average $=70$ (2).	1+2+2

	OR: Pure part in the first $= 300 \times \frac{48}{100} = 144$ mL (1); Pure part in the second $= 255$ mL (1); Total $= 399$ mL in 1050 mL (1); Concentration $= 38\%$ (2).	1+1+1+2
20	Totals: 2484, 2350, 4136 (1); Grand total $= 8970$ for 130 students (2); Average $= 69$ (2).	1+2+2
	OR: Pure part in the first $= 800 \times \frac{39}{100} = 312$ mL (1); Pure part in the second $= 66$ mL (1); Total $= 378$ mL in 1350 mL (1); Concentration $= 28\%$ (2).	1+1+1+2

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